In [ ]:
```
 Experiment 3: Creating DataFrames & Analyzing Properties
PIN:-241UDS0097

 Aim :-
Create Pandas DataFrames using NumPy arrays and a CSV file, then inspect
DataFrame properties such as shape, index, columns, data types, information,
descriptive statistics, and transpose.
```

In [ ]:
```
## Step 1: Install and Import Libraries
Run the install command only if NumPy or Pandas is missing from your
environment.
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2: Create a 2-D NumPy Array
Rows represent charging stations and columns represent the first four weeks.
```

In [2]:
```
charging_sessions_241UDS0097 = np.array([
    [142, 157, 169, 181],
    [188, 201, 194, 223],
    [121, 139, 151, 164]
])

charging_sessions_241UDS0097
```

Out[2]:
```
array([[142, 157, 169, 181],
       [188, 201, 194, 223],
       [121, 139, 151, 164]])
```

In [3]:
```
# Check array properties: shape and number of dimensions
charging_sessions_241UDS0097.shape, charging_sessions_241UDS0097.ndim
```

Out[3]:
```
((3, 4), 2)
```

In [ ]:
```
## Step 3: Convert the 2-D Array into a DataFrame
Set station names as the index and week labels as columns, then add Week 5
and a fourth station.
```

In [4]:
```
charging_index_241UDS0097 = [
    "Charging_Station_A",
    "Charging_Station_B",
    "Charging_Station_C"
]

charging_columns_241UDS0097 = ["Week_1", "Week_2", "Week_3", "Week_4"]

charging_df_241UDS0097 = pd.DataFrame(
    data=charging_sessions_241UDS0097,
    index=charging_index_241UDS0097,
    columns=charging_columns_241UDS0097
)

charging_df_241UDS0097
```

Out[4]:

	Week_1	Week_2	Week_3	Week_4
Charging_Station_A	142	157	169	181
Charging_Station_B	188	201	194	223
Charging_Station_C	121	139	151	164

In [5]:
```
# Add Week 5 data (one value per existing station)
charging_df_241UDS0097["Week_5"] = [193, 231, 172]
charging_df_241UDS0097
```

Out[5]:

	Week_1	Week_2	Week_3	Week_4	Week_5
Charging_Station_A	142	157	169	181	193
Charging_Station_B	188	201	194	223	231
Charging_Station_C	121	139	151	164	172

In [6]:
```
# Add a fourth charging station as a new row
station_d_241UDS0097 = pd.DataFrame(
    {
        "Week_1": [166],
        "Week_2": [179],
        "Week_3": [187],
        "Week_4": [205],
        "Week_5": [216]
    },
    index=["Charging_Station_D"]
)

charging_df_241UDS0097 = pd.concat(
    [charging_df_241UDS0097, station_d_241UDS0097]
)
charging_df_241UDS0097
```

Out[6]:

	Week_1	Week_2	Week_3	Week_4	Week_5
Charging_Station_A	142	157	169	181	193
Charging_Station_B	188	201	194	223	231
Charging_Station_C	121	139	151	164	172
Charging_Station_D	166	179	187	205	216

In [ ]:
```
## Step 4: Create a DataFrame from an Existing CSV File
Keep the CSV in the same folder as this notebook. The CSV includes the
original three stations' first four weeks; the DataFrame created above
demonstrates adding Week 5 and another station.
```

In [7]:
```
charging_csv_df_241UDS0097 = pd.read_csv(
    "Experiment_3_241UDS0097_Electric_Vehicle_Charging.csv",
    index_col=0
)

charging_csv_df_241UDS0097
```

Out[7]:

Charging_Station	Week_1	Week_2	Week_3	Week_4
Charging_Station_A	142	157	169	181
Charging_Station_B	188	201	194	223
Charging_Station_C	121	139	151	164

In [ ]:
```
## Step 5: Analyze Important DataFrame Properties
Inspect the first rows, dimensions, index, data types, and column names.
```

In [ ]:
```
charging_csv_df_241UDS0097.head()
```

In [ ]:
```
charging_csv_df_241UDS0097.shape
```

In [ ]:
```
charging_csv_df_241UDS0097.index
```

In [ ]:
```
charging_csv_df_241UDS0097.dtypes
```

In [ ]:
```
charging_csv_df_241UDS0097.columns
```

In [ ]:
```
## Step 6: Information About the DataFrame
```

In [ ]:
```
charging_csv_df_241UDS0097.info()
```

In [ ]:
```
## Step 7: Descriptive Statistics
Summarize the numeric week columns using count, mean, standard deviation,
minimum, quartiles, and maximum.
```

In [ ]:
```
charging_csv_df_241UDS0097.describe()
```

In [ ]:
```
## Step 8: Transpose Rows and Columns
Transpose the first few rows so the week labels and station records switch
orientation.
```

In [ ]:
```
charging_csv_df_241UDS0097.head(10).transpose()
```

In [ ]:
```
## Result
A DataFrame was created from a NumPy array and from a CSV file. DataFrame
rows and columns were added, and the shape, index, data types, column names,
information, descriptive statistics, and transposed view were examined.
```